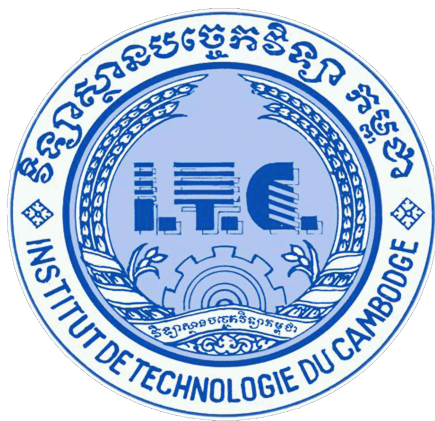




Name: Virak Rith

Student ID: P20230033

Course: Networks System Design

Instructor: KUY Movsun

Assignment: Lab9(Network Layer Data Plane)

Due Date: December 30, 2025 (12:00 AM)

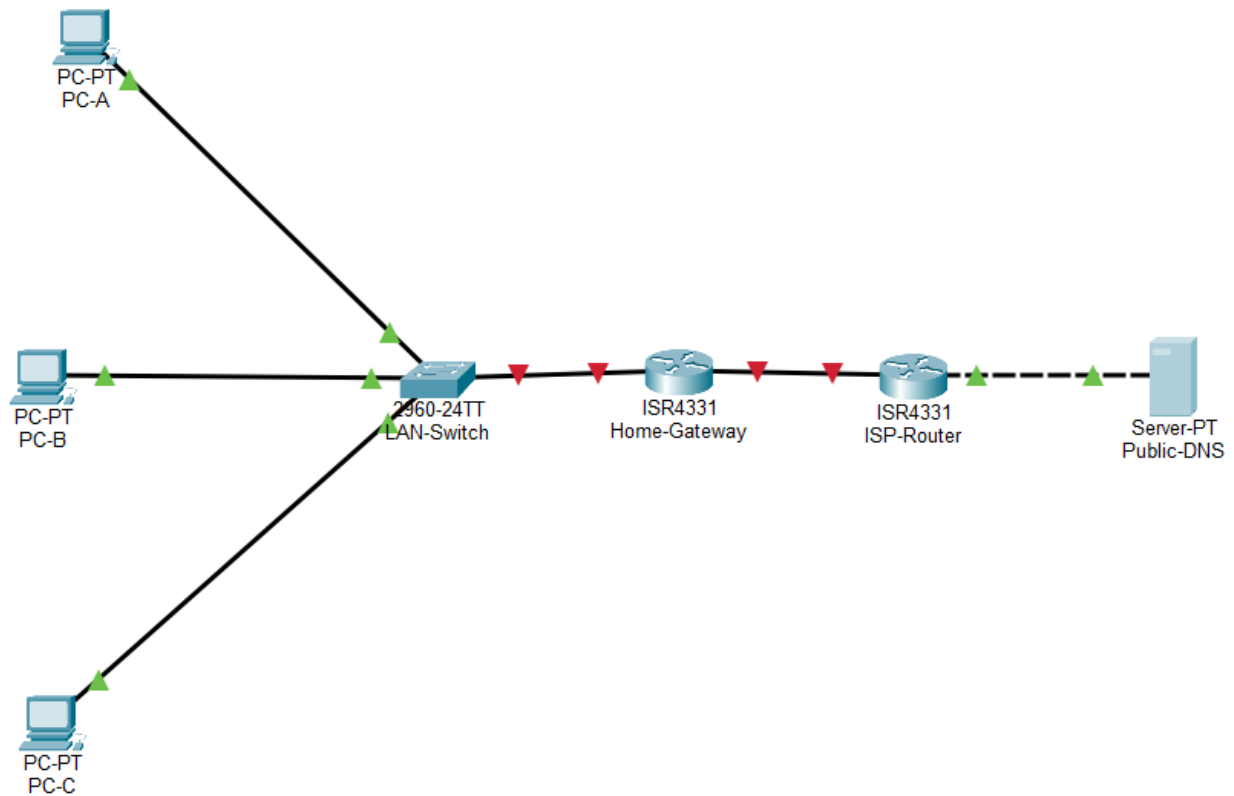
Part 1: Topology & Addressing

Device Table

Device Name	Model	Interface	Description
ISP-Router	ISR 4331	Gig0/0/0	WAN Link (to Home-Gateway)
		Gig0/0/1	Internet Link (to Server)
Home-Gateway	ISR 4331	Gig0/0/0	WAN Uplink (to ISP)
		Gig0/0/1	LAN Gateway (to Switch)
LAN-Switch	2960	Any	Layer 2 Switch
Public-DNS	Server-PT	NIC	Simulated Internet Server (8.8.8.8)

Cabling Guide

- ISP-Router (G0/0/0) ↔ Home-Gateway (G0/0/0): Copper Crossover (or Straight-through if Auto-MDIX)
- ISP-Router (G0/0/1) ↔ Public-DNS: Copper Crossover
- Home-Gateway (G0/0/1) ↔ LAN-Switch (G0/1): Copper Straight-through
- LAN-Switch ↔ PC-A, PC-B, PC-C: Copper Straight-through



Part 2: Pre-Lab Setup (Instructor Configuration)

Configure Public-DNS Server

1. Go to **Public-DNS > Desktop > IP Configuration**
2. Set:
 - IP Address: **8.8.8.8**
 - Subnet Mask: **255.255.255.0**
 - Default Gateway: **8.8.8.1**

Configure ISP-Router (CLI)

```
Router> enable
Router# configure terminal
Router(config)# hostname ISP-Router

! Connection to Server
Router(config)# interface g0/0/1
Router(config-if)# ip address 8.8.8.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

! Connection to Home-Gateway
Router(config)# interface g0/0/0
Router(config-if)# ip address 203.0.113.1 255.255.255.252
```

```
Router(config-if)# no shutdown
Router(config-if)# end
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ISP-Router
ISP-Router(config)# interface g0/0/1
ISP-Router(config-if)#ip address 8.8.8.1 255.255.255.0
ISP-Router(config-if)#no shutdown

ISP-Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
exit
ISP-Router(config)#exit
ISP-Router#
%SYS-5-CONFIG_I: Configured from console by console

ISP-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ISP-Router(config)#interface g0/0/0
ISP-Router(config-if)#ip address 203.0.113.1 255.255.255.252
ISP-Router(config-if)#no shutdown

ISP-Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

ISP-Router(config-if)#end
ISP-Router#
%SYS-5-CONFIG_I: Configured from console by console

ISP-Router#
```

Task 3: Implementing NAT (Network Address Translation)

Scenario

Convert the LAN to a private network (**192.168.1.0/24**) and configure NAT Overload (PAT) so internal hosts can access the Internet.

Step 3A: Configure WAN Link & Default Route

```
Home-Gateway(config)# interface g0/0/0
Home-Gateway(config-if)# description WAN-Link
Home-Gateway(config-if)# ip address 203.0.113.2 255.255.255.252
Home-Gateway(config-if)# no shutdown
Home-Gateway(config-if)# exit

! Default route to ISP
Home-Gateway(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

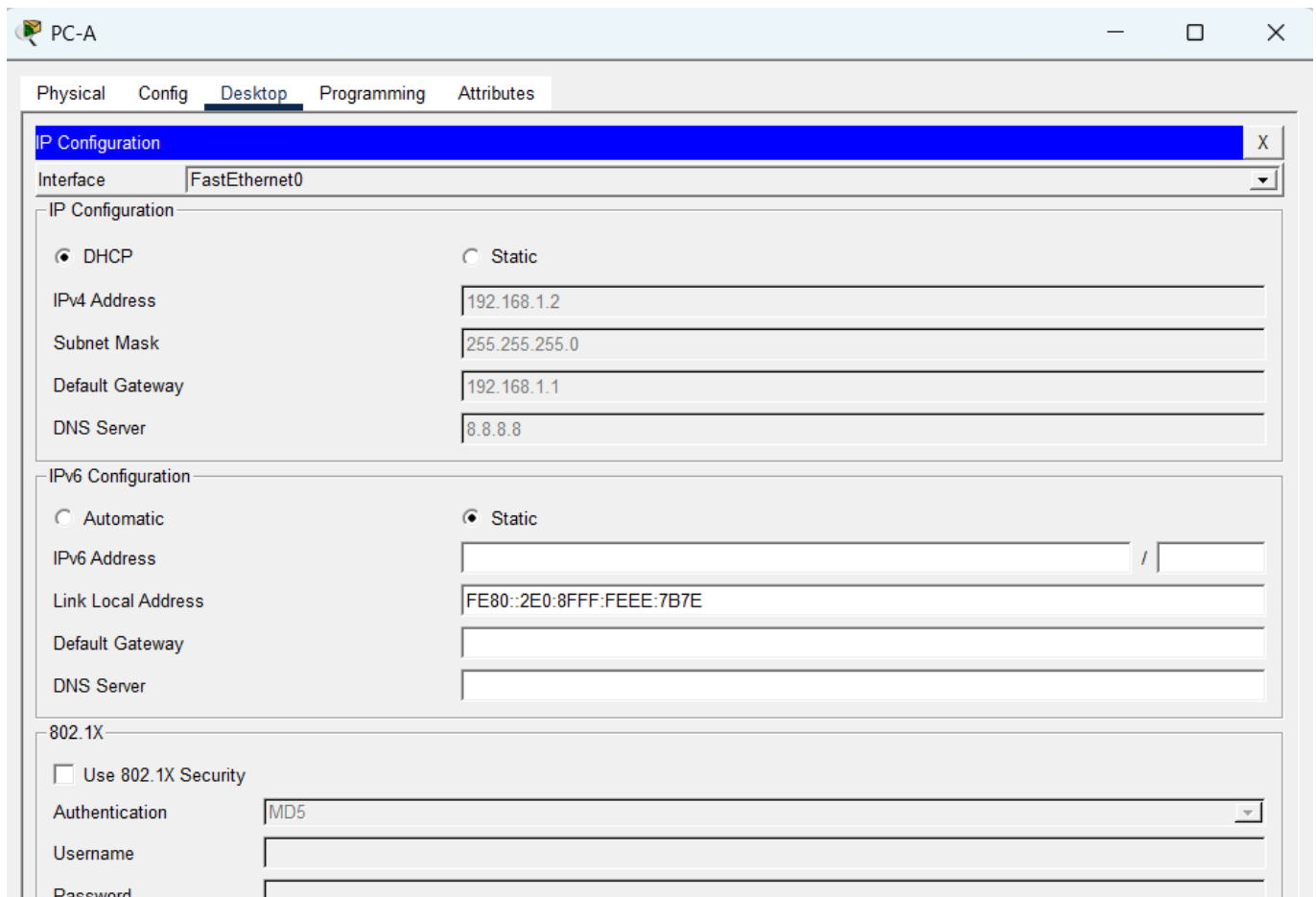
Step 3B: Re-configure LAN as Private Network

```
! Change LAN Gateway IP
Home-Gateway(config)# interface g0/0/1
Home-Gateway(config-if)# ip address 192.168.1.1 255.255.255.0
Home-Gateway(config-if)# exit

! Update DHCP Pool
Home-Gateway(config)# no ip dhcp pool LAN_POOL
Home-Gateway(config)# ip dhcp pool PRIVATE_LAN
Home-Gateway(dhcp-config)# network 192.168.1.0 255.255.255.0
Home-Gateway(dhcp-config)# default-router 192.168.1.1
Home-Gateway(dhcp-config)# dns-server 8.8.8.8
Home-Gateway(dhcp-config)# exit
```

Step 3C: Force PCs to renew IP

On PC-A, PC-B, and PC-C: Open the IP Configuration window, switch to Static, wait a second, then switch back to DHCP. Verify they now have an address like 192.168.1.2.



Step 3D: Configure NAT Overload (PAT)

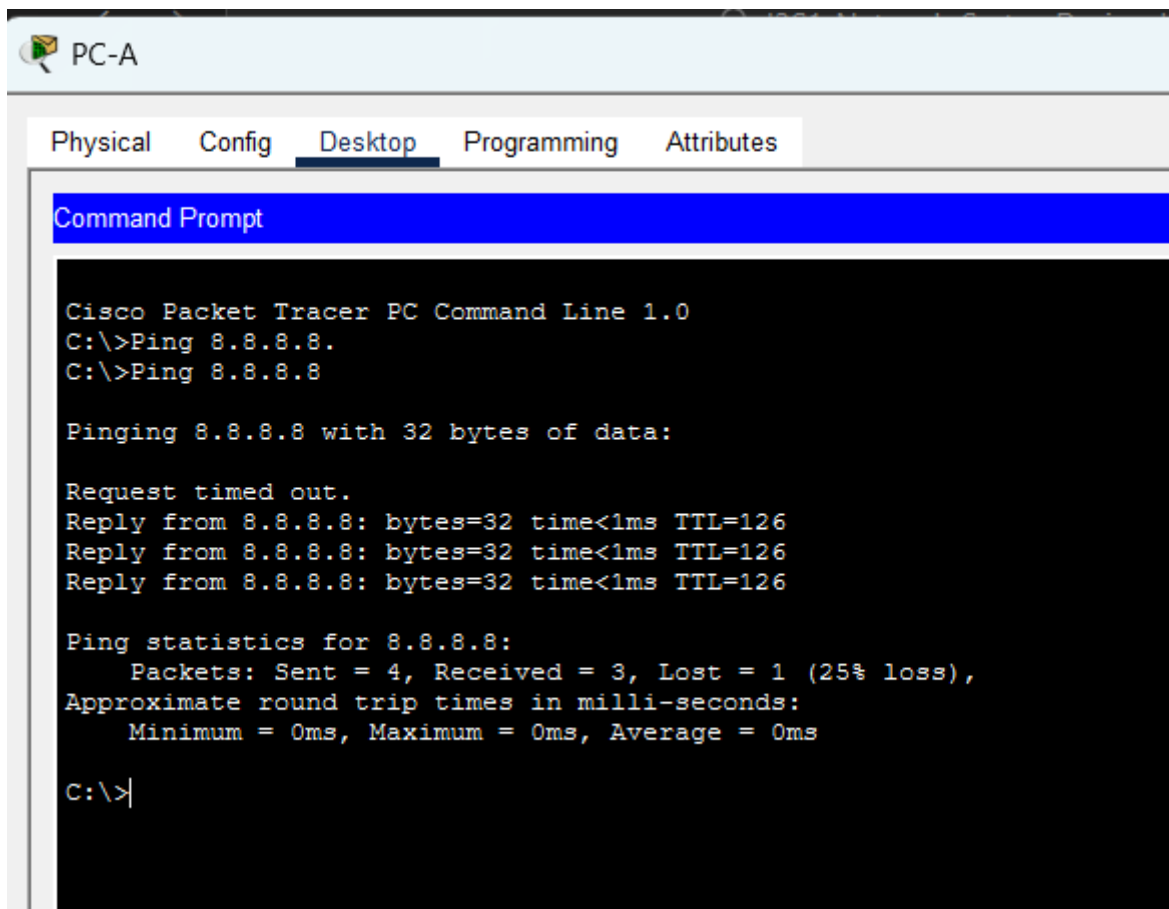
```
! Define Inside and Outside interfaces
Home-Gateway(config)# interface g0/0/1
Home-Gateway(config-if)# ip nat inside
Home-Gateway(config-if)# exit
```

```
Home-Gateway(config)# interface g0/0/0
Home-Gateway(config-if)# ip nat outside
Home-Gateway(config-if)# exit

! Create ACL for LAN traffic
Home-Gateway(config)# access-list 1 permit 192.168.1.0 0.0.0.255

! Apply NAT Overload
Home-Gateway(config)# ip nat inside source list 1 interface g0/0/0 overload
```

- Verification: Open Command Prompt on PC-A. Ping 8.8.8.8. If configured correctly, the ping will succeed.



- Check the router table: show ip nat translations.

```
Home-Gateway#
%SYS-5-CONFIG_I: Configured from console by console

Home-Gateway#show ip nat translation
Pro Inside global      Inside local      Outside local      Outside global
udp 203.0.113.2:1025    192.168.1.2:1025  8.8.8.8:53         8.8.8.8:53

Home-Gateway#
```

```
! Enable IPv6 routing
Home-Gateway(config)# ipv6 unicast-routing
```

```
! Configure IPv6 on LAN interface
Home-Gateway(config)# interface g0/0/1
Home-Gateway(config-if)# ipv6 address 2001:DB8:ACAD:1::1/64
Home-Gateway(config-if)# no shutdown
Home-Gateway(config-if)# exit
```

- Verification: On PC-C > IP Configuration, look at the "IPv6 Configuration" section. Ensure it is set to Auto Config. It should automatically generate an IPv6 address.

The screenshot shows the configuration window for PC-C, with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing the 'FastEthernet0' interface. The 'IP Configuration' section has two radio buttons: 'DHCP' (selected) and 'Static'. The 'DHCP' section shows a successful request and the following values: IPv4 Address: 192.168.1.4, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1, and DNS Server: 8.8.8.8. The 'IPv6 Configuration' section also has two radio buttons: 'Automatic' (selected) and 'Static'. The 'Automatic' section shows a successful request and the following values: IPv6 Address: 2001:DB8:ACAD:1:230:F2FF:FE70:7AE3 / 64, Link Local Address: FE80::230:F2FF:FE70:7AE3, Default Gateway: FE80::260:2FFF:FE44:CA02, and DNS Server: (empty). The '802.1X' section is also expanded, showing a checkbox for 'Use 802.1X Security' (unchecked), a dropdown for 'Authentication' set to 'MD5', and empty fields for 'Username' and 'Password'.

PC-C

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.1.4

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 8.8.8.8

IPv6 Configuration

☒ Automatic ☐ Static Ipv6 request successful.

IPv6 Address 2001:DB8:ACAD:1:230:F2FF:FE70:7AE3 / 64

Link Local Address FE80::230:F2FF:FE70:7AE3

Default Gateway FE80::260:2FFF:FE44:CA02

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password